

Nalen Rangarajan

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Olathe, KS

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EDUCATION

Kansas State University

May 2026

B.S. in Computer Science; B.S. in Mathematics; 4.0 GPA; Summa Cum Laude

Manhattan, KS

- Relevant coursework: Network Programming, Programming Language Design, Computer Architecture and Operations, Algorithm Analysis, Software Architecture and Design, Database System Concepts, Object-Oriented Programming, Data and Programming Structures
- Recipient of the 2026 Outstanding Senior for Excellence in Mathematics Award
- Founding member of K-State's Cube Satellite program
- William Lowell Putnam Mathematical Competition. Score: 1, 2024
- Inducted into Tau Beta Pi (Engineering Honor Society, top 12.5% of Juniors)
- Studied abroad in Tokyo and Kyoto, Japan

WORK EXPERIENCE

Garmin

January 2025 – August 2025, Summer 2026

Software Engineering Intern

Manhattan and Olathe, KS

Worked across the project execution phase for integrated flight deck applications, from UI specifications and software requirement design to software development and DO-178B verification. Collaborated with product design and software teams to develop integrated GUI software in C for the G3000 and AXIS Flight Deck families.

- **Designed** 3 unique GUI interfaces for displaying database data for an upcoming application for AXIS
- **Implemented** a redesign of the G3000 window manager application which included adding support for landscape orientation, failure modes, and general QoL improvements
 - **Improved** automatic pane selection for multi-unit display configurations
 - **Authored** software requirements for new functionalities from internal stakeholders
 - **Verified** application functionality and reliability through requirements-based module and unit tests

Kansas State CubeSat

January 2025 – May 2026

Commands and Data Handling Subsystem Lead

Manhattan, KS

- **Formulated** in-depth data budgets for payload and telemetry data for the PHAT-Sat Imaging and NuKat Radiation Detection missions
- **Conducted** frequency band and data rate trade studies to inform high-level flight software design decisions
- **Presented** data budget and flight software design information to industry reviewers in system requirement reviews (SRR) and preliminary design reviews (PDR)
- **Developed** subsystem requirements and risk evaluations for C&DH to ensure mission compliance

Kansas State Mathematics Department

January 2025 – May 2026

Undergraduate Research Assistant

Manhattan, KS

- **Implemented** a dominating set algorithm to find a minimum-weight maximal increasing subsequence in a permutation using Python
- **Implemented** a dual active set algorithm to solve strictly convex quadratic programs using Python
- **Presented** at the K-State Research Symposium and K-State Directed Reading Program Conferences

Kansas State Computer Science Department

December 2023 – January 2026

Undergraduate Research Assistant

Manhattan, KS

- **Developed** a formally verified knowledge graph reasoner using Coq and Ocaml
- **Optimized** a linear programming algorithm to process various predicates showcasing a 70-400x speedup in reasoner performance
- **Presented** at the 2025 Joint Mathematics Meetings in Seattle, WA

- **Taught** students fundamental data structures and algorithms in C#, including stacks, queues, hash tables, recursion, searching, and tree traversal
- **Assisted** students one-on-one and in small groups during office hours with in-class content, lab exercises, and homework projects

PUBLICATIONS AND PRESENTATIONS

- Atalay Mert Ileri, **Nalen Rangarajan**, Jack Cannell, Hande Küçük McGinty. *VEL: A Formally Verified Reasoner for OWL2 EL Profile*. arXiv:2412.08739 (preprint), 2024.
- **Nalen Rangarajan**, Atalay Mert Ileri, Jack Cannell, Hande Küçük McGinty. (Jan 2025). *The Quest for VE-EL: A Formally Verified and Enhanced EL++ Reasoner*. Poster presented at the 2025 **Joint Mathematics Meetings**, Seattle, WA, USA. (Solo presenter).
- **Nalen Rangarajan**. (May 2025). *Modulus of families of maximal increasing subsequences*. Presentation at the Spring 2025 Kansas State Directed Reading Program Conference, Manhattan, KS, USA.
- **Nalen Rangarajan**. (Dec 2025). *Expectations of modulus of families of maximal increasing subsequences*. Presentation at the Fall 2025 Kansas State Directed Reading Program Conference, Manhattan, KS, USA.
- **Nalen Rangarajan**. (May 2026). *A Dual Active Set Algorithm – Goldfarb-Idnani*. Presentation at the Spring 2026 Kansas State Directed Reading Program Conference, Manhattan, KS, USA.
- Vinny Sun, **Nalen Rangarajan**. (Apr 2025). *Modulus of Permutations*. Poster presented at the Kansas State 2025 Undergraduate Research Symposium, Manhattan, KS, USA.

PERSONAL PROJECTS

- **Dataying Block Programming** – Developed a full stack application utilizing Vue and C# designed to teach young students data science concepts through a drag-and-drop programming style
 - Created a custom lesson creation and management environment for teachers
- **Video Game Database** – Developed an interactive video game database application using SQL and C#
- **Chat Application** – Developed an application for messaging utilizing C# and websockets

SKILLS

- **Programming Languages:** C#, C, Vue, Coq, Ocaml, MSSQL, Python, Java, XML, HTML
- **Frameworks and Libraries:** .NET, RESTful APIs, Windows Forms, WPF, Razor Pages
- **Tools and Version Control:** Git, JIRA, Visual Studio, VS Code, Bitbucket, Confluence, Unity
- **Math:** Mathematical Analysis, Complex Analysis, Data & Network Mathematics, Calculus III, Linear Algebra, Discrete Math, Differential Equations, Statistics, Algebraic Systems